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Effects of step-down weaning implementation time on growth performance and blood metabolites of dairy calves

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ABSTRACT

The current study evaluated the effects of step-down weaning implementation time on starter feed intake, growth performance, blood metabolites, and ruminal pH in dairy calves. A total of 48 Holstein dairy calves (24 male and 24 female; 3 ± 1 d old; 41.2 ± 1.8 kg of body weight) were assigned (n = 12 per treatment; 6 male and 6 female) to 4 experimental treatments in a completely randomized block design. All calves were fed 6 L/d of milk from d 3 to 10 of age, and the treatments consisted of the following: calves were offered 8 L/d of milk from d 11 to 28 of age and then 4 L/d from d 29 to 63 (Step-28; total milk offered = 326 L); calves were offered 8 L/d of milk from d 11 to 42 of age and then 4 L/d from d 43 to 63 (Step-42; total milk offered = 382 L); calves were offered 8 L/d of milk from d 11 to 56 of age and 4 L/d from d 57 to 63 (Step-56; total milk intake = 438 L); and calves were fed 8 L/d of milk from d 11 to 63 of age and abruptly weaned (control; total milk offered = 466 L). All calves were housed individually in pens and had ad libitum access to water and solid feed throughout the experiment. All calves were completely milk weaned on d 64, and their performance was measured until d 80 of age. During the experiment, the starter intake (kg/d and % of body weight) was greater in calves in the Step-28 group compared with those in the other groups. However, the total metabolizable energy intake was greater in the Step-56 calves compared with the other calves. Overall, the average daily gain (ADG), feed efficiency (ADG/dry matter intake), and ADG/total metabolizable energy intake were similar across the treatments. Circulating glucose, β-hydroxybutyrate, blood urea nitrogen, albumin, total protein, alanine aminotransferase, and aspartate aminotransferase, as well as ruminal pH, were not affected by the treatments. The implementation of step-down weaning in early life (4–6 wk of age) could stimulate solid feed intake compared with weaning at a later age with no negative effect on performance.

Key words: liquid feed, step-up/step-down, weaning, dairy calf

INTRODUCTION

Early-life nutrition and management of dairy calves could influence their development and growth during the first few months of life, which dictates their future performance in dairy herds. After colostrum feeding, milk allowance and weaning management are the most critical factors dictating the growth and health of artificially reared calves (Khan et al., 2011; Mirzaei et al., 2018). Research during the last 2 decades has clearly shown the benefits of feeding higher milk allowance (8 L/d or more) to promote organ development, growth, and welfare of artificially reared heifer calves (Eckert et al., 2015; Geiger et al., 2016; Khan et al., 2016). Furthermore, published literature regarding the positive relationship between preweaning weight gain and future milk yield (Soberon and Van Amburgh, 2013; Chester-Jones et al., 2017) has promoted the use of higher milk feeding to rear Heifer replacement calves for commercial dairy herds (Welboren et al., 2019). However, higher milk allowance has been associated with depressed solid feed intake, which delays ruminal development; therefore, the transition from milk to solid feed becomes challenging for calves (Khan et al., 2016).

Calves fed a high milk allowance generally undergo a period of depressed solid feed intake around weaning, which results in depressed growth over several weeks after milk weaning (Khan et al., 2011, 2016). Research to develop management strategies to promote solid feed consumption in calves fed higher levels of milk is therefore required to maintain the growth advantages

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after milk weaning. Several strategies can be used to stimulate the starter intake during the milk feeding period to ease the transition from milk to solid feed for calves fed higher milk allowance. Calf age at weaning, method of weaning, social housing, and solid feeds enriched with nutrients (Costa et al., 2016; Berends et al., 2018; Welboren et al., 2019) have been tested to promote solid feed intake in calves fed higher levels of milk. Out of these, gradual weaning—namely, the step-down method—at 4 wk of calf age has been proposed (Khan et al., 2007) to promote solid feed intake before weaning and to maintain BW gain during and after weaning. Another procedure, called the step-up/step-down method, has been proposed by Omidi-Mirzaei et al. (2015) to provide greater amounts of milk to calves to encourage starter intake and promote ruminal development and growth performance. Others have proposed a linear reduction of milk allowance between 5 to 12 wk (Sweeney et al., 2010; de Passillé and Rushen, 2016; Welboren et al., 2019). However, implementation of weaning through linear or step-down procedures at an early age could still result in decrease in growth of calves (Khan et al., 2007; Eckert et al., 2015; McCoard et al., 2019), and implementation of such procedures is difficult to adopt in many commercial farms where calves are still being fed manually.

Therefore, in the present study, it was hypothesized that delaying the age of implementation of a step-down procedure beyond 4 wk of age will promote solid feed intake and therefore help minimize the negative effects of milk weaning on the growth performance of calves fed manually. The objective of this study was to investigate the effects of age at the time of implementing step-down weaning on feed intake, growth performance, blood metabolites, and ruminal fermentation in dairy calves.

MATERIALS AND METHODS

Animals, Treatments, and Management

This study was performed from October 8, 2018, to January 18, 2019, at Foudeh Agriculture and Animal Husbandry Co., Isfahan, Iran. All the animal procedures were approved by the Animal Care and Use Committee of Arak University (IACUC #201801IR) by the Iranian Council of Animal Care.

A total of 48 Holstein dairy calves (41.2 ± 1.8 kg of initial BW; 24 male and 24 female) were used in the study. Immediately after birth, calves were separated from their dams, weighed, and moved to individual pens (1.2 × 2.5 m) bedded with sand, which was renewed every 24 h. Calves consumed 3.5 L of colostrum within 1 h of life and 3.5 L of colostrum 9 h after the first feeding. The quality of colostrum was measured with a digital Brix refractometer (PAL-1, Atago Co. Ltd., Bellevue, WA) on d 2 of life, calves received transition milk (4 L) in 2 meals of equal volume (at 0900 and 1900 h), and then healthy calves (with no clinical symptoms of diarrhea or systemic diseases) were randomly assigned to experimental treatments (12 calves per treatment; 6 male and 6 female) in a completely randomized block design.

From study d 1 onward (d 3 of age), all calves were fed 6 L/d of milk from d 3 to 10 and treatments were as follows (Figure 1): calves were offered 8 L/d of milk from d 11 to 28 of age and then 4 L/d from d 29 to 63 (**Step-28**; total milk offered = 344 L); calves were offered 8 L/d of milk from d 11 to 42 of age and then 4 L/d from d 43 to 63 (**Step-42**; total milk offered = 400 L); calves were offered 8 L/d of milk from d 11 to 56 of age and 4 L/d from d 57 to 63 (**Step-56**; total milk intake = 456 L); and calves were fed 8 L/d of milk from d 11 to 63 of age and abruptly weaned (**CON**; total milk offered = 487 L). All calves were fed pasteurized whole milk (3.18 ± 0.1% fat, 3.08 ± 0.08% CP, 4.56 ± 0.06% lactose, and 11.40 ± 0.15% TS) in steel buckets in 2 meals of equal volume per day (at 0800 and 1600 h), and the study was terminated on d 80 of age. Calf health was checked daily by a veterinarian blinded to the treatments, and sick calves were treated by a veterinarian as required.

Starter feed with ground physical form (Table 1) was fed ad libitum to permit at least 10% orts (i.e., the portion of the starter not consumed over a 24-h period). Starter feed refusals were collected and recorded at 0800 h, and fresh starter feed was fed at 0830 h. All calves had free access to clean drinking water from a bucket throughout the experimental period.

Sampling and Analysis

Milk samples were collected once per week in the morning and the afternoon of the same day for analysis. The milk samples were preserved using bronopol and stored at 8°C until analysis. Samples were analyzed for lactose, fat, CP, and TS content by MilkoScan (Foss Electric, Hillerød, Denmark; AOAC International, 1997). Throughout the study, daily feed offered and refused was weighed to determine the total starter intake for the individual calf. Samples of starter feed were collected throughout the study (n = 7, 1 sample per 10 d) for the determination of DM and chemical analyses. Subsamples of dried feed were ground in a mill (Ogaw Seiki Co. Ltd., Tokyo, Japan) to pass through a 1-mm screen before chemical analysis. Feed samples were

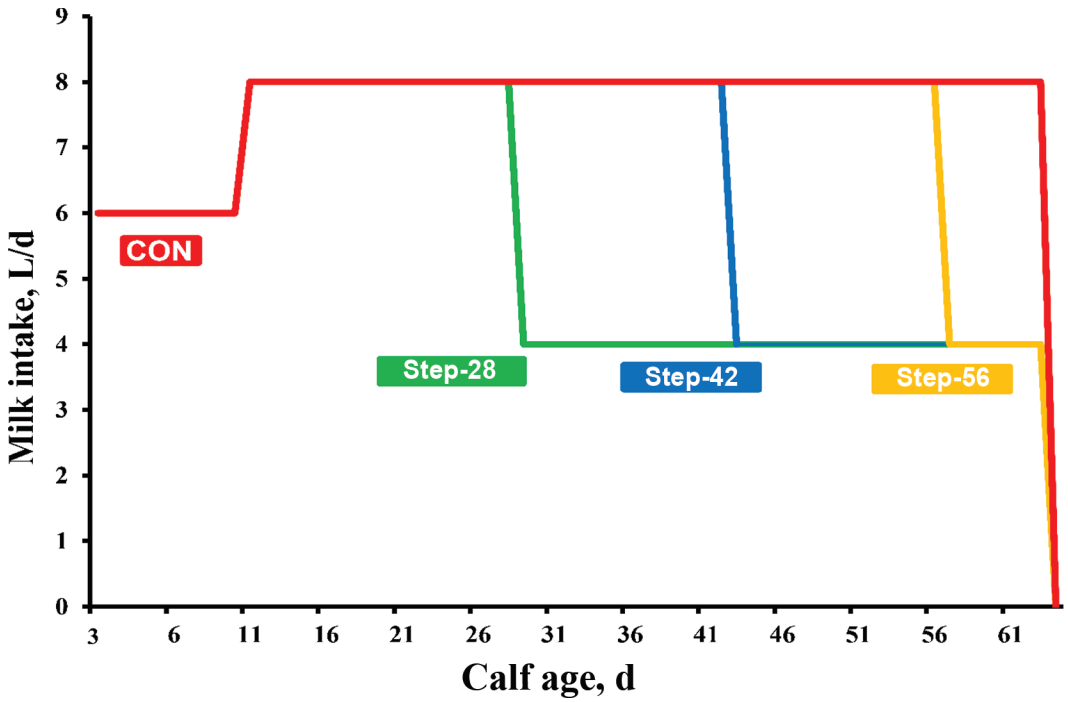

Figure 1. The milk intake (L/d) of calves on the different milk feeding programs. Step-28 = calves were offered 8 L/d of milk from d 11 to 28 of age and then 4 L/d from d 29 to 63; total milk offered = 326 L. Step-42 = calves were offered 8 L/d of milk from d 11 to 42 of age and then 4 L/d from d 43 to 63; total milk offered = 382 L. Step-56 = calves were offered 8 L/d of milk from d 11 to 56 of age and 4 L/d from d 57 to 63; total milk intake = 438 L. CON (control) = calves were fed 8 L/d of milk from d 11 to 63 of age; total milk offered = 466 L.

Table 1. Ingredients and chemical composition (% of DM unless otherwise noted) of the basal starter feed

Item	Value
Ingredient	
Alfalfa hay, chopped	10.0
Corn grain, ground	45.0
Barley grain, ground	14.0
Soybean meal, 45.5% CP	26.5
Energizer RP-10[1]	1.25
Vitamin and mineral mix[2]	0.75
Sodium bicarbonate	0.75
Calcium carbonate	1.25
Salt	0.5
Chemical composition	
ME,[3] Mcal/kg	3.27
NE_G,[3] Mcal/kg	1.56
DM	90.0
CP	19.2
Ether extract	4.4
NDF	14.0
NFC[4]	54.5
Ca	0.9
P	0.4

[1]Prilled saturated free fatty acids (IFFCO, Johor Bahru, Malaysia).

[2]Contained per kilogram of supplement: vitamin A, 1,500,000 IU; vitamin D, 50,000 IU; vitamin E, 1,000 IU; Ca, 100 g; P, 20 g; Mg, 40 g; Zn, 2,000 mg; Cu, 500 mg; I, 100 mg; Co, 120 mg; Mn, 1,500 mg; Se, 100 mg.

[3]Calculated according to NRC (2001).

[4]Calculated as DM − (NDF + CP + ether extract + ash) (NRC, 2001).

analyzed for CP (AOAC, 1990; method 988.05), NDF without the use of sodium sulfite and with the inclusion of α-amylase (Van Soest et al., 1991), and ADF (AOAC, 1990; method 973.18). The NFC component was calculated as 100 − (CP + NDF + ether extract + ash) (NRC, 2001).

Calves were weighed every 10 d and ADG (kg of BW/d) was calculated. Feed efficiency was computed as kilograms of ADG divided by kilograms of total DMI (liquid feed DMI + starter feed DMI). Body measurements including body length (distance between the point of shoulder and rump), body girth (distance around the belly over the part of the back before morning feeding), withers height (distance from the base of the front feet to the withers), heart girth (circumference of the chest), hip height (distance from the base of the rear feet to hook bones), and hip width (distance between the points of hook bones) of the calves were recorded on d 3 of calf age, at weaning (d 64 of calf age), and on d 80 of age according to the method described by Khan et al. (2007).

Blood samples were collected from the jugular vein into 10-mL serum tubes at 4 h after morning feeding on d 28, 42, 56, 63, and d 77. Blood samples were placed on ice immediately after collection and within 20 min were centrifuged at 3,000 × g for 15 min at 4°C

to obtain serum. The serum was preserved at −20°C until analysis to determine concentrations of glucose (kit no. 93008), albumin (kit no. 9307), total protein (kit no. 9304), and BUN (kit no. 93013) using commercial kits according to the manufacturer's instructions (Pars Azmoon Co., Tehran, Iran). Plasma BHB was determined with the autoanalyzer using a laboratory kit (Randox Laboratories Ltd., Ardmore, UK). Liver function indicator enzymes, aspartate aminotransferase (**AST**), and alanine aminotransferase (**ALT**) were measured using ELISA (717 AutoAnalyzer, Hitachi, Tokyo, Japan) and commercial laboratory kits (kits no. 92005 and 92004 for AST and ALT, respectively; Pars Azmoon Co.). On d 35 and 70 of age, ruminal fluid was sampled by a stomach tube fitted to a vacuum pump 4 h after morning feeding. The pH was immediately measured using a hand-held pH meter (HI 8318, Hanna Instruments, Cluj-Napoca, Romania).

Statistical Analysis

Power analysis for sample size estimation was performed (Morris, 1999; Hintze, 2008) for the primary response variables, including feed intake, BW, and ADG based on previously published values (DeVries and von Keyserlingk, 2009; Zhang et al., 2010; Miller-Cushon and DeVries, 2011). From the power test analysis, using $\alpha = 0.05$ and power = 0.80, the projected sample size was 12 calves per treatment for DMI, ADG, and hip height. Data were subjected to ANOVA using the MIXED procedure of SAS 9.4 (SAS Institute Inc., Cary, NC) with time as repeated measures (each period was 10 d) for starter feed intake, DMI, ME intake, ADG, BW, feed efficiency, skeletal growth, ruminal pH, and blood metabolites during the overall (d 3–80 of age) period. The model consisted of treatment, time (sampling date), treatment × time interaction, and sex as the fixed effects and calf as a random effect. Before analyses, all data were screened for normality using the UNIVARIATE procedure of SAS. Three variance–covariance structures (autoregressive type 1 and compound symmetry) were tested, and the covariance structure that minimized Schwarz's Bayesian information criterion was chosen. For BW and structural growth variables, the initial measurements were included as a covariate for the statistical analysis of future data. Least squares means for treatment effects were separated by the use of PDIFF. The threshold of significance was set at $P \leq 0.05$; trends were declared at $0.05 < P \leq 0.10$.

RESULTS

Intake and Growth Performance

Intake and growth performance data are presented in Table 2. The CON calves consumed more ($P < 0.01$) milk (466 L of the total 487 L of milk offered) than the Step-28 (326 L of the total 344 L of milk offered), Step-42 (382 L of the total 400 L of milk offered), and Step-56 (438 L of the total 456 L of milk offered) calves. A higher intake of starter feed (Figure 2A) was observed for the calves weaned using the Step-28 method compared with the CON group during the overall period ($P < 0.01$), with no differences between the Step-42 and Step-56 methods. The intake of starter feed as a percentage of BW was highest for calves weaned using

Table 2. Effects of age at the implementation of step-down weaning on feed intake, growth, and feed efficiency in calves during the entire experiment (d 3–80 of age, n = 12 per treatment)

	Treatment[1]					P-value[2]		
Variable	Step-28	Step-42	Step-56	CON	SEM	Trt	Per	Trt × Per
Starter intake, g/d	1,065[a]	908[ab]	971[a]	760[b]	48.6	<0.01	<0.01	0.23
Starter intake, % of BW	1.28[a]	1.05[b]	1.06[b]	0.83[c]	0.06	<0.01	<0.01	0.03
Milk DMI, g/d	692[d]	808[c]	925[b]	962[a]	2.0	<0.01	<0.01	<0.01
Total DMI,[3] g/d	1,603	1,533	1,683	1,519	52.2	0.09	<0.01	0.49
Total ME intake, Mcal/d	6.38[b]	6.33[b]	7.01[a]	6.56[ab]	0.16	0.02	<0.01	0.02
ADG, g/d	812	760	805	799	38.6	0.76	<0.01	0.46
Feed efficiency[4]	0.51	0.50	0.48	0.53	0.02	0.75	<0.01	0.69
ADG/total ME intake	0.13	0.12	0.12	0.12	0.005	0.34	<0.01	0.69

[a–d]Means within a row with different superscript letters are different ($P < 0.05$).

[1]Step-28 = calves were offered 8 L/d of milk from d 11 to 28 of age and then 4 L/d from d 29 to 63; total milk offered = 326 L. Step-42 = calves were offered 8 L/d of milk from d 11 to 42 of age and then 4 L/d from d 43 to 63; total milk offered = 382 L. Step-56 = calves were offered 8 L/d of milk from d 11 to 56 of age and 4 L/d from d 57 to 63; total milk intake = 438 L. CON (control) = calves were fed 8 L/d of milk from d 11 to 63 of age; total milk offered = 466 L.

[2]Trt = effect of treatment; Per = effect of period; Trt × Per = interaction between treatment and period.

[3]DMI = starter DM + milk DM intakes.

[4]Feed efficiency was calculated by dividing ADG (g) by average daily DMI.

the Step-28 method compared with calves weaned later during the overall period ($P < 0.01$).

The total DMI tended ($P = 0.09$) to be smaller for the CON group compared with the other groups. The total ME intake was greatest ($P = 0.02$) for the Step-56 group compared with the other groups. The age of the step-down implementation did not affect ADG (Figure 2B), feed efficiency, or ADG/total ME intake during the overall period. No differences were observed among the weaning methods regarding BW, structural growth parameters, and ruminal pH during the overall period (Table 3). The intakes of starter feed as a percentage of BW ($P = 0.03$), milk DM, ($P < 0.01$), and total ME ($P = 0.02$) were also affected by the interaction between age of the step-down implementation and time. However, we observed no interaction between the age of

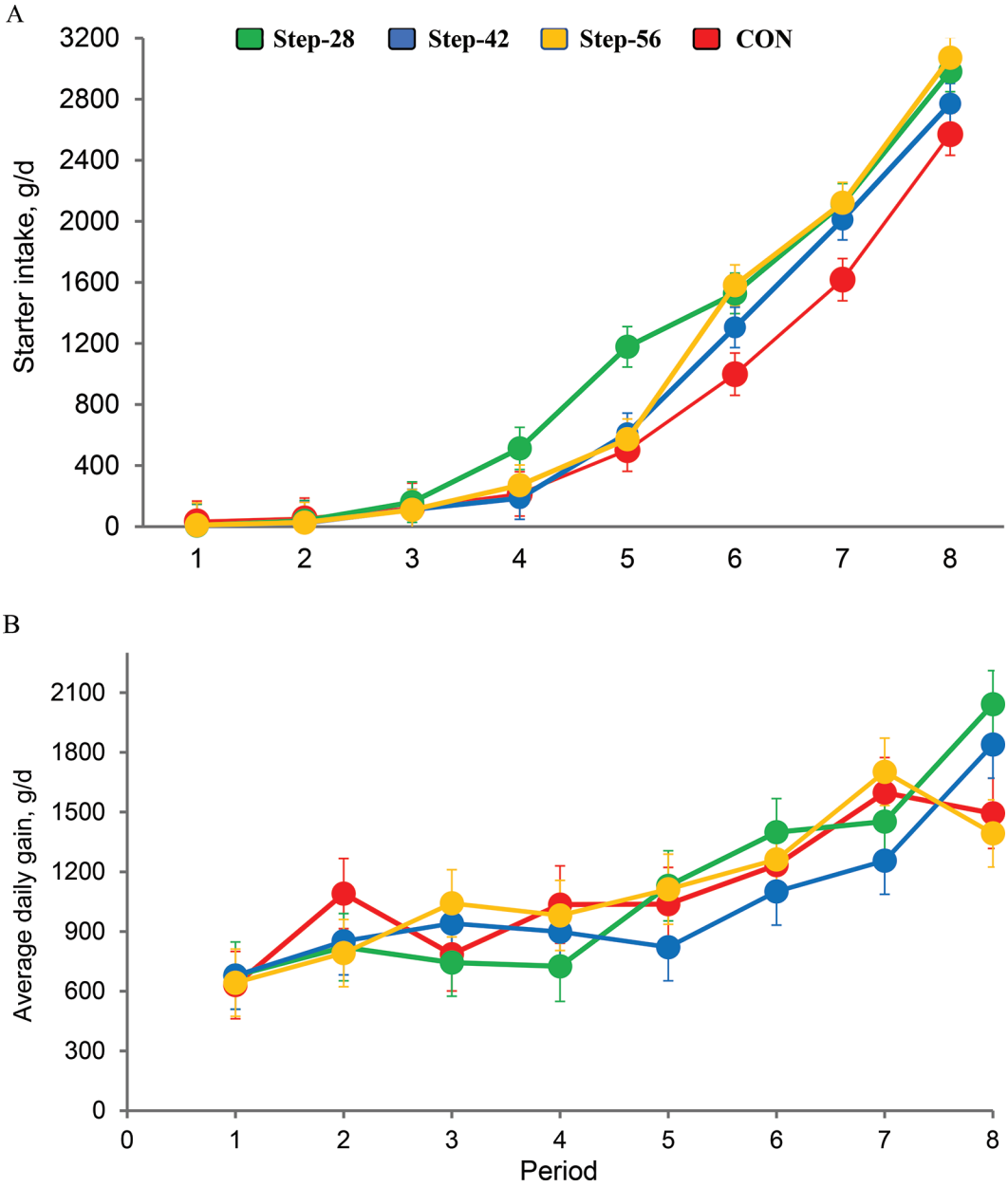

Figure 2. The starter feed intake (A) and ADG (B) of calves on the different milk feeding programs. Step-28 = calves were offered 8 L/d of milk from d 11 to 28 of age and then 4 L/d from d 29 to 63; total milk offered = 326 L. Step-42 = calves were offered 8 L/d of milk from d 11 to 42 of age and then 4 L/d from d 43 to 63; total milk offered = 382 L. Step-56 = calves were offered 8 L/d of milk from d 11 to 56 of age and 4 L/d from d 57 to 63; total milk intake = 438 L. CON (control) = calves were fed 8 L/d of milk from d 11 to 63 of age; total milk offered = 466 L.

Table 3. Effects of age at the implementation of step-down weaning on growth characteristics of dairy calves (n = 12 per treatment)

	Treatment[1]					*P*-value[2]		
Variable	Step-28	Step-42	Step-56	CON	SEM	Trt	Per	Trt × Per
BW, kg	72.4	70.1	72.0	71.7	1.4	0.54	<0.01	0.79
Heart girth, cm	104.4	106.2	106.1	105.9	1.3	0.74	<0.01	0.52
Wither height, cm	90.7	92.7	92.8	92.1	1.1	0.47	<0.01	0.88
Body length, cm	58.6	58.4	61.8	60.6	2.0	0.61	<0.01	0.11
Body barrel, cm	122.6	121.7	124.6	121.4	1.8	0.61	<0.01	0.99
Hip height, cm	96.8	97.7	97.6	97.3	0.8	0.84	0.03	0.45
Hip width, cm	29.7	31.0	30.4	32.0	0.8	0.21	<0.01	0.69
Ruminal pH	5.85	6.06	6.06	6.12	0.13	0.54	0.07	0.19

[1]Step-28 = calves were offered 8 L/d of milk from d 11 to 28 of age and then 4 L/d from d 29 to 63; total milk offered = 326 L. Step-42 = calves were offered 8 L/d of milk from d 11 to 42 of age and then 4 L/d from d 43 to 63; total milk offered = 382 L. Step-56 = calves were offered 8 L/d of milk from d 11 to 56 of age and 4 L/d from d 57 to 63; total milk intake = 438 L. CON (control) = calves were fed 8 L/d of milk from d 11 to 63 of age; total milk offered = 466 L.

[2]Trt = effect of treatment; Per = effect of period; Trt × Per = interaction between treatment and period.

the step-down implementation and time on the skeletal growth parameters (Table 3; Supplemental Table S1, https://doi.org/10.3168/jds.2020-18520).

Blood Metabolites and Ruminal pH

No differences in the serum concentrations of glucose, BHB, BUN, albumin, total protein, AST, and ALT (Table 4) were observed among the calves weaned at different ages. As calves aged, the serum concentrations of glucose decreased but the serum concentrations of BHB ($P < 0.01$), BUN ($P < 0.01$), albumin ($P < 0.01$), total protein ($P = 0.02$), AST ($P < 0.01$), and ALT ($P < 0.01$) increased (Table 4). The serum concentrations of BHB were also affected by the interaction between age of the step-down implementation and time (tendency, $P = 0.06$): calves weaned using the Step-28 method had greater BHB compared with the calves weaned using the other methods on d 42 (Supplemental Table S2, https://doi.org/10.3168/jds.2020-18520).

DISCUSSION

Calves on commercial dairy farms are generally weaned with respect to age or certain target BW (Vasseur et al., 2010; Shivley et al., 2018). Weaning transition from liquid feed to solid feed is a stressful event, especially for calves that are fed on a high plane of milk or milk replacer (Jasper et al., 2008). To offset the weaning transition challenge, a weaning protocol called "step-down" was proposed by Khan et al. (2007) to create a gradual weaning process starting around calf age of 4 wk. Subsequent studies (Sweeney et al., 2010; Eckert et al., 2015; Welboren et al., 2019) have argued

Table 4. Effects of age at the implementation of step-down weaning on ruminal pH and blood metabolites of dairy calves (n = 12 per treatment)

	Treatment[2]					*P*-value[3]		
Variable[1]	Step-28	Step-42	Step-56	CON	SEM	Trt	Per	Trt × Per
Glucose, mg/dL	110.1	108.3	116.6	119.7	5.92	0.18	<0.01	0.56
BHB, mmol/L	0.23	0.20	0.20	0.19	0.02	0.13	<0.01	0.06
BUN, mg/dL	10.2	10.9	10.6	10.2	0.33	0.42	<0.01	0.56
AST, U/L	39.9	41.4	37.4	40.0	2.37	0.69	<0.01	0.35
ALT, U/L	9.6	8.8	12.4	10.3	1.30	0.47	<0.01	0.21
Total protein, g/dL	7.0	6.8	6.9	6.8	0.08	0.13	0.02	0.75
Albumin, g/dL	3.5	3.5	3.6	3.5	0.03	0.84	<0.01	0.80

[1]AST = aminotransferase; ALT = alanine aminotransferase.

[2]Step-28 = calves were offered 8 L/d of milk from d 11 to 28 of age and then 4 L/d from d 29 to 63; total milk offered = 326 L. Step-42 = calves were offered 8 L/d of milk from d 11 to 42 of age and then 4 L/d from d 43 to 63; total milk offered = 382 L. Step-56 = calves were offered 8 L/d of milk from d 11 to 56 of age and 4 L/d from d 57 to 63; total milk intake = 438 L. CON (control) = calves were fed 8 L/d of milk from d 11 to 63 of age; total milk offered = 466 L.

[3]Trt = effect of treatment; Per = effect of period; Trt × Per = interaction between treatment and period.

that implementation of linear or step-down procedures at an early age could still result in decrease in growth of calves around weaning because early weaning creates a lag between the demand and supply of energy and nutrients needed to maintain a high growth rate in calves. Therefore, in an attempt to refine the implementation of step-down procedures, this study was conducted to test the hypothesis that calves would perform better by delaying the age of implementation of a step-down procedure beyond 4 wk of age, which could promote solid feed intake and therefore help minimize the negative effects of milk weaning on growth performance.

It was found that early implementation of a step-down weaning procedure triggered the initiation of starter intakes at an early age (d 28 and 42) compared with implementation of step-down weaning at later ages (d 56) or abrupt weaning at d 63. Moreover, the calves that were stepped down at d 28 and 42 of age did not show any weight loss after weaning, and their BW gain from d 3 to 63 of age was comparable with that of the calves that were stepped down at d 56 or abruptly weaned at d 63 of age. From d 64 to 80 of age, the Step-28 calves were able to compensate for the reduced milk through elevated starter feed intake and thus did not experience low growth. The Step-42 calves partly compensated for the low milk allowance by increasing the intake of starter feed; however, they did not gain enough weight after wk 4 and during the postweaning stage (d 64–80 of age) to compensate fully for the lower allowance of milk compared with the Step-28 calves. Previous studies have demonstrated that gradual weaning from milk can improve intake of starter feed and lead to improved energy intakes during the weaning transition when fed with high rates of milk replacer (Khan et al., 2007; Sweeney et al., 2010). Dennis et al. (2018) concluded that feeding calves with high milk replacer allowance (1.1 kg of DM/d) and weaning them gradually over 2 wk at d 53 resulted in similar BW and growth compared with calves weaned at d 42 of age. However, Hill et al. (2012) compared gradual weaning at d 42 of age and reported greater weight gain and BW from d 43 to 56 in calves that were gradually weaned. Recently, Bittar et al. (2020) reported that the adoption of gradual weaning is not effective in improving calf performance when calves receive a low amount of milk before weaning (3 wk), but it decreases vocalization on d 2 after weaning. In summary and in contrast to our initial hypothesis, once the step-down implementation began, calves at age 28 d quickly increased their starter feed intake so that the overall ME intake and BW gains during the preweaning period were similar to those of calves that stepped down at the late age. Preweaning growth has been associated with the mammary gland development and future lactation performance of heifers (Gelsinger et al., 2016). In the present study, similar preweaning BW gains in heifer calves weaned at different ages indicated that early implementation of a step-down procedure may not have any negative influence on the future performance of heifer replacements; however, this suggestion needs to be evaluated in future research.

Normal physiological concentrations and lack of differences in the serum concentrations of glucose, BHB, BUN, albumin, total protein, AST, and ALT as well as ruminal pH in the calves weaned using a step-down procedure implemented at different ages indicated that weaning age did not affect the major circulating metabolites associated with nutrient metabolism and liver functions. In summary, implementation of a step-down procedure at an early age has triggered the initiation of solid feed intake and therefore metabolic development of the ruminal epithelium earlier in Step-28 calves compared with calves weaned at a later age.

In the current study, we investigated the weaning strategies for calves that were individually housed and had received a high daily milk allowance that was manually delivered in 2 feeding frequencies. In the absence of automatic feeders, feeding calves twice per day is still the most common method of milk feeding in commercial dairy farms (Vasseur et al., 2010; Shivley et al., 2018). However, several studies (Sweeney et al., 2010; de Passillé and Rushen, 2016; Benetton et al., 2019) investigated gradual weaning protocols using automated milk feeding systems that allow calves to express their natural feeding behavior and provide them with an opportunity for social learning when housed in groups (Costa et al., 2015). Providing calves with the opportunity to express natural behaviors and mimic natural suckling behavior along with social learning from peers in group-housed calves could further help trigger the early initiation of solid feed consumption. Recent literature (Lecorps et al., 2018; Neave et al., 2018) has suggested that the weaning age of calves is also influenced by their individual characteristics and behavioral traits (e.g., learning and personality), implying that individual traits should be taken into account to further refine the weaning protocols to ease the transition of calves from milk to solid feed. However, more investigations using automatic feeding systems and sensing technologies are needed to develop individualized feeding and weaning protocols for artificial rearing of calves on commercial farms. Development and adaptation of an automatic milk feeding system are undergoing a transitory phase, and most of the commercial farms across the globe are still using a manual system, delivering the milk twice a day to calves. Therefore, it is concluded that implementation of a step-down weaning procedure at an early age (4–6 wk) will promote solid feed intake within the

preweaning period to ease the weaning transition for calves manually offered higher milk allowance.

CONCLUSIONS

The implementation of a step-down procedure earlier (at 4 wk of calf age) is a better strategy than implementing gradual weaning at a later age to promote solid feed intake and to minimize the loss in BW gain around weaning in individually housed calves that are fed milk with 2 feeding frequencies. Studies using a large number of replacement heifers are warranted to evaluate the effects of age of implementing a step-down weaning procedure on mammary gland development, future growth, and lactation performance.

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